MERCY

St Teresa's School

- Founded by the Sisters of Mercy in 1930 -

# Tour de Maths 2024

## Leg 2

29 February 2024

Question Paper

**Sponsored by:**

CASIO

## 5 MARK QUESTIONS

**QUESTION 1**

What is the smallest positive integer that is a multiple of four, six and ten?

**QUESTION 2**

$2021 - 2020 + 2019 - 2018 + \cdots + 5 - 4 + 3 - 2 + 1$ equals

**QUESTION 3**

The time $2021$ minutes after $20:21$ on a digital 24-hour clock is

**QUESTION 4**

If January 1st 2019 was a Tuesday, how many Tuesdays were there in 2019?

**QUESTION 5**

The positive integers are written in a long sequence $12345678910111213$ … When the sequence contains 100 digits, how many of them are 1's?

**QUESTION 6**

Your sister will be $x$ years old in the year $x^2$ . In what year was your sister born?

**QUESTION 7**

After how many hours will the second candle be twice the height of the first candle, given that both were lit simultaneously and one burns out in 3 hours while the other takes 4 hours to burn completely?

**QUESTION 8**

What is the tens digit of N = 1 x 3 x 5 x 7 x ⋯ x 99?

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**QUESTION 9**

Determine the number of possible distinct pairs of integers $(a; b)$ for which

$\left(2a + 5b - \frac{1}{2}\right)^2 \leq 2 - (a + 2b)^2$

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**QUESTION 10**

Find the value of $p + q$ if $p, q \in \mathbb{Z}^+$ and $p + q + pq = 54$.

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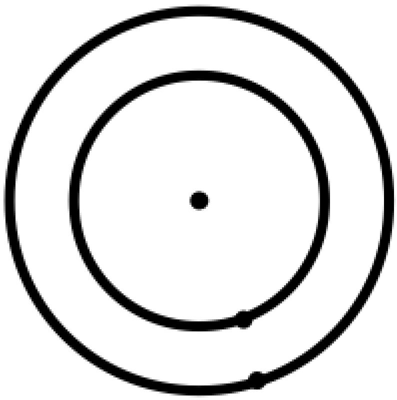
## 10 MARK QUESTIONS

### QUESTION 11

John, Simon and Peter decide to paint a wall. John could have painted it himself in 180 minutes, while Simon would have taken 4 hours and working together John, Simon and Peter can paint it in 1 hour 20 minutes. How long would it take the Peter to paint the wall alone?

### QUESTION 12

The diagram shows two concentric circles. If the circumference of one exceeds the other by $6\ cm$ by how much does its radius exceed the other radius?



### QUESTION 13

Determine the number of pairs $(a; b)$ of integer solutions for $2^{2a} = 55 + 3^{3b}$

**QUESTION 14**

The function $\gamma(x)$ is defined for all $x \epsilon \mathbb{R}$. If $\gamma(a+b) = \gamma(a)\gamma(b)$ for all $a, b$ and $\gamma(1) = 2$.

Find the value of $\gamma(2013)$.

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**QUESTION 15**

Find the number of subsets of $X = \{1,2,3,4,5,6,7,8\}$ that are subsets of neither $Y = \{1,2,3,4,5\}$, nor $Z = \{4,5,6,7,8\}$

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## 20 MARK QUESTIONS

### QUESTION 16

Between $12:00$ and $13:00$, there are two times when the hands of a clock are exactly at right angles. How many minutes apart are these two times?


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### QUESTION 17

Neither of the two integers $x$ and $y$ is divisible by 3. What are the possible remainders when $x^3 + y^3$ is divided by 9?

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### QUESTION 18

From the numbers $1, 2, 3, \ldots, 500$, a sequence is formed by deleting numbers so that no two of the remaining numbers have a sum which is a multiple of 7.

What is the maximum number of numbers in this sequence?

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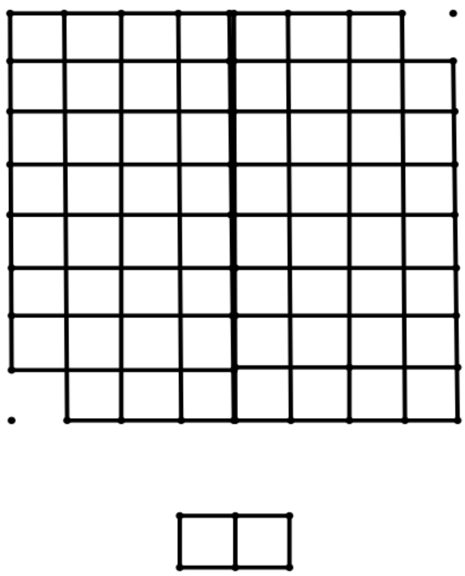
**QUESTION 19**

Two opposite corners of an $8 \times 8$ grid are removed, so 62 squares are left.
You have thirty-one $1 \times 2$ dominos. Place the dominos on the grid in such a way that all 62 squares on the grid are occupied.



**QUESTION 20**

A point P has distances 3, 4 and 5 from the vertices of an equilateral triangle, as shown.
Calculate the length of one side of the equilateral triangle.